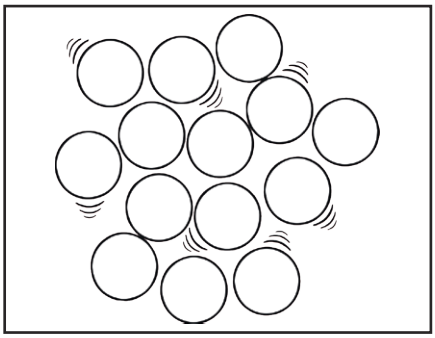
Name:	Class:	Date given:
		Date due in:

Particulate Nature **of Matter**

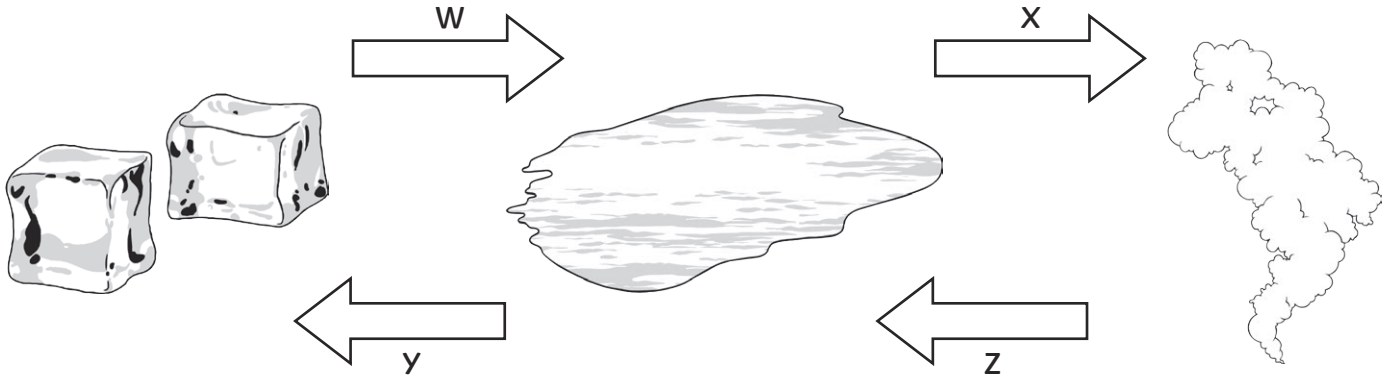
1. Complete the table by ticking or crossing the properties for each state of matter. [6]

Property	Solid	Liquid	Gas
High density			
Low density			
Fixed shape			
Fixed volume			
Easily squashed			
Flows			

2. Complete the particle diagrams for solid and gas states of matter. [2]



3. Look at the diagram.
   State the process which is occurring at each of the points labelled W, X, Y and Z. [4]



W: _____ X: _____

Y: _____ Z: _____

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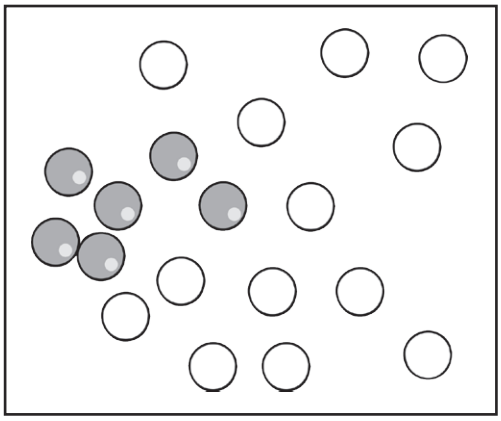
Susie blows up a balloon. First, she places the balloon in a bucket of ice water. She observes the balloon. She then removes the balloon and places it into a bucket of hot water. She observes the balloon again.

4a. Explain what Susie will observe when she puts the balloon in ice water. [3]

4b. Explain what Susie will observe when she puts the balloon in hot water. [3]

5. What is diffusion? [2]

6. Draw particles in the second box to show the smell after it has diffused into the air. [1]



Learning Outcomes (tick if achieved)

Q1	I can identify properties of SLG	
Q2	I can draw particle arrangement of SLG	
Q3	I can recall changes of state processes	
Q4	I can describe how particles change with changing state	
Q5	I can describe diffusion	

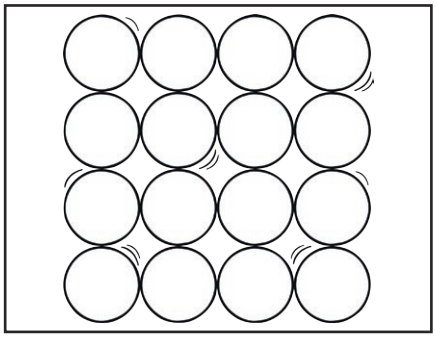
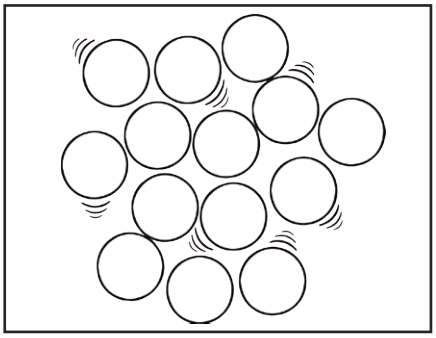
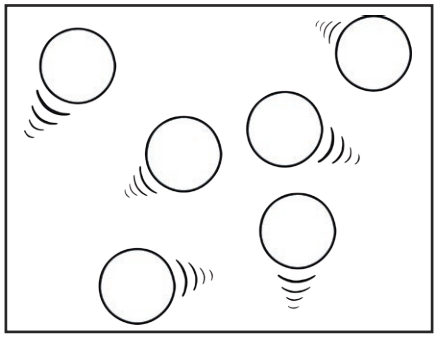
Particulate Nature of Matter **Answers**

1. Complete the table by ticking or crossing the properties for each state of matter. [6]

Property	Solid	Liquid	Gas
High density	✓	✗	✗
Low density	✗	✗	✓
Fixed shape	✓	✗	✗
Fixed volume	✓	✓	✗
Easily squashed	✗	✗	✓
Flows	✗	✓	✓

1 mark for each correct row

2. Complete the particle diagrams for solid and gas states of matter. [2]

3. Look at the diagram.
State the process which is occurring at each of the points labelled W, X, Y and Z. [4]

W: **melting**

X: **evaporating**

Y: **freezing**

Z: **condensing**

4a. Explain what Susie will observe when she puts the balloon in ice water. [3]

balloon will shrink / get smaller

because air particles condense

air particles have less energy

move less / collide with balloon wall less

decreased air pressure

4b. Explain what Susie will observe when she puts the balloon in hot water. [3]

balloon will expand / get bigger

because air particles spread out

air particles have more energy

move more / collide with balloon wall more

increased air pressure

5. What is diffusion? [2]

spreading out of particles

to reach equilibrium

2 marks for:

movement from area of high concentration to area of low concentration

6. Draw particles in the second box to show the smell after it has diffused into the air. [1]

